Bounded Count Regression Models

Binomial, beta-binomial and beta-2-binomial

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1. Introduction: Motivation and Scope

- Bounded count responses have support $Y \in \{0, 1, \ldots, m\}$.
- Examples: health scores, symptoms, correct answers, and grouped binary responses.
- Binomial regression is the baseline for bounded counts.
- BB regression adds overdispersion through δ.
- B2B regression adds shape flexibility through ν (Bayes et al. 2024).
- The analysis and figures are reproducible with `BoundedCountRegression.Rmd`.

1.1 Examples of Bounded Count Data

Bounded count regression can be fitted when Y is a count with a known maximum m:

- health scores: number of symptoms, adverse events, or positive health items out of a fixed checklist (Bayes et al. 2024);
- educational testing: number of correct answers out of m items (Johnson et al. 2005);
- grouped binary responses: number of successes among m trials, patients, plots, or experimental units (Johnson et al. 2005);
- clinical and epidemiological scales: bounded severity or function scores with excess zeros or upper-bound observations (Bayes et al. 2024);
- quality-control studies: number of defective components in a fixed batch (Johnson et al. 2005).

These data often require models beyond the binomial when the variance, tails, or boundary counts are not adequately represented.

2. Bounded Count Distributions: Data Structure

For a fixed maximum count m,

$$Y = 0, 1, \ldots, m.$$

The value of m is known by design:

- number of trials,
- number of test items,
- maximum score,
- maximum observable events.

2.1 Bounded Count Distributions: Binomial and BB

Binomial model:

$$Y \sim \text{Binomial}(m, \mu), \qquad \Pr(Y = y) = \binom{m}{y} \mu^y (1-\mu)^{m-y}.$$

$$\mathbb{E}(Y) = m\mu, \qquad \text{Var}(Y) = m\mu(1-\mu).$$

Beta-Binomial model:

$$Y \mid \pi \sim \text{Binomial}(m, \pi), \qquad \pi \sim \text{Beta}(\alpha, \beta).$$

$$\Pr(Y = y) = \binom{m}{y} \frac{B(y+\alpha, m-y+\beta)}{B(\alpha, \beta)}.$$

Mean-dispersion form, with $\mu = \alpha/(\alpha+\beta)$ and $\delta = 1/(\alpha+\beta+1)$:

$$\mathbb{E}(Y) = m\mu, \quad \text{Var}(Y) = m\mu(1-\mu)\{1 + (m-1)\delta\}.$$

2.2 Bounded Count Distributions: Beta-2-Binomial

$$Y \mid W = w \sim \text{BetaBinomial}(m, \mu, w), \qquad W \sim \text{Beta}(\delta, \nu).$$

$$\text{b2b}(y \mid m, \mu, \delta, \nu) = \int_0^1 \text{bb}(y \mid m, \mu, w)\, \text{be}(w \mid \delta, \nu)\, dw.$$

Here Beta uses mean–dispersion parameters, not two shape parameters:

$$a = \delta(1-\nu)/\nu, \qquad b = (1-\delta)(1-\nu)/\nu.$$

The B2B model preserves the BB mean and variance, while ν changes higher-order shape (Bayes et al. 2024).

2.3 Parameter Interpretation

$$\theta = (\mu) \quad \text{or} \quad \theta = (\mu, \delta) \quad \text{or} \quad \theta = (\mu, \delta, \nu).$$

- $\mu \in (0, 1)$ is the success probability or mean proportion.
- $\mathbb{E}(Y) = m\mu$, so μ is directly interpretable on the count scale through $m\mu$.
- $\delta \in (0, 1)$ controls extra-binomial variation in BB and B2B models.
- $\nu \in (0, 1)$ controls higher-order shape in the B2B model.
- The B2B parameter ν can alter tails and boundary probabilities while preserving the BB mean-variance form.

2.4 Distributional Hierarchy

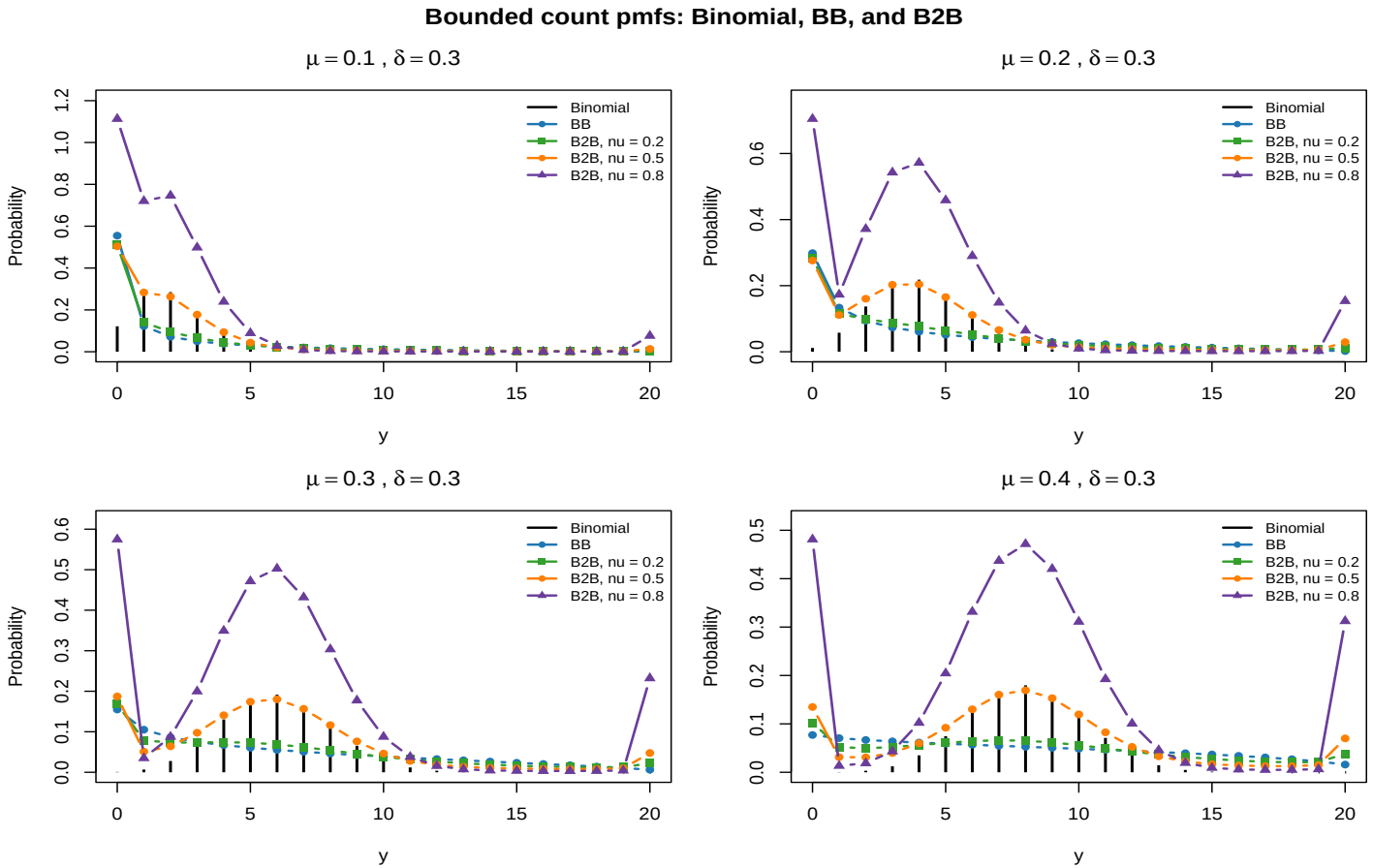
$$\text{Binomial} \subset \text{BetaBinomial} \subset \text{Beta2Binomial}\,.$$

$$\delta \to 0 \quad \Rightarrow \quad \text{BB} = \text{Binomial}\,.$$

$$\nu \to 0 \quad \Rightarrow \quad \text{B2B} = \text{BB}\,.$$

These inclusions hold through boundary limits (Bayes et al. 2024).

2.5 Figure 1: Distributional Shapes



Adapted from Figure 2 in Bayes et al. (2024); see the manuscript for settings.

2.6 Identification and Monotonicity

- Binomial: $\mu = \mathbb{E}(Y)/m$.
- BB: μ and δ are identified by the first two moments when $m > 1$.
- B2B: ν requires information beyond the first two moments; small m can prevent identification.
- Binomial is monotone in μ.
- BB is monotone in μ for fixed δ.
- B2B is monotone in μ for fixed δ and ν.
- If dispersion or shape vary jointly with μ, stochastic ordering is not guaranteed.

2.7 Skewness and Kurtosis: Shape Functions

Let

$$q = \mu(1-\mu), \qquad A_x = 1 + (m-1)x.$$

For $Y \sim \text{Binomial}(m, \mu)$,

$$\text{Skew}_{Bin} = \frac{1-2\mu}{\sqrt{mq}}, \qquad \text{Kurt}_{Bin} = 3 + \frac{1-6q}{mq}.$$

Kurtosis here is the standardized fourth central moment, not excess kurtosis.

BB and B2B preserve mean structure but modify skewness and kurtosis through δ and ν.

2.8 Theorem 1: Skewness Relationships

For $Y \sim \text{BetaBinomial}(m, \mu, \delta)$,

$$\text{Skew}_{BB}(m, \mu, \delta) = K(m, \delta)\,\text{Skew}_{Bin}(m, \mu), \quad K(m, \delta) = \frac{1 + (2m-1)\delta}{(1+\delta)\sqrt{A_\delta}}.$$

For $Y \sim \text{B2B}(m, \mu, \delta, \nu)$ and $W \sim \text{Beta}(\delta, \nu)$,

$$\text{Skew}_{B2B}(m, \mu, \delta, \nu) = \frac{\mathbb{E}\left[\text{Skew}_{BB}(m, \mu, W) A_W^{3/2}\right]}{A_\delta^{3/2}}.$$

Special cases:

$$\delta \to 0 \Rightarrow \text{Skew}_{BB} = \text{Skew}_{Bin}, \qquad \nu \to 0 \Rightarrow \text{Skew}_{B2B} = \text{Skew}_{BB}\,.$$

2.9 Theorem 2: Kurtosis Relationships

For $Y \sim \text{BetaBinomial}(m, \mu, \delta)$,

$$\text{Kurt}_{BB}(m, \mu, \delta) = \frac{mq\ \text{Kurt}_{Bin}(m, \mu) + k_1\delta + k_2\delta^2 + k_3\delta^3}{(1+\delta)(1+2\delta)mqA_\delta^2}.$$

For $Y \sim \text{B2B}(m, \mu, \delta, \nu)$ and $W \sim \text{Beta}(\delta, \nu)$,

$$\text{Kurt}_{B2B}(m, \mu, \delta, \nu) = \frac{\mathbb{E}\left[\text{Kurt}_{BB}(m, \mu, W)A_W^2\right]}{A_\delta^2}.$$

Special cases:

$$\delta \to 0 \Rightarrow \text{Kurt}_{BB} = \text{Kurt}_{Bin}, \qquad \nu \to 0 \Rightarrow \text{Kurt}_{B2B} = \text{Kurt}_{BB}\,.$$

2.9 Theorem 2: Coefficient Definitions

The coefficients in Theorem 2 are, with $q = \mu(1-\mu)$,

$$
\begin{aligned}
k_0 &= 3q(m-2)+1, \\
k_1 &= 3q(2m^2-11m+6)+7m-4, \\
k_2 &= 3q(m^3-16m^2+19m-6) \\
&\quad +12m^2-15m+5, \\
k_3 &= 3q(m-1)(-7m^2+7m-2) \\
&\quad +2(m-1)(3m^2-3m+1).
\end{aligned}
$$

The accompanying manuscript gives the proofs of both moment relationships.

2.10 Shape Parameters

- Binomial skewness changes sign at $\mu = 0.5$.
- BB keeps the same skewness sign as Binomial, but δ changes magnitude.
- BB kurtosis contains the binomial kurtosis plus δ-dependent terms.
- B2B skewness and kurtosis use weighted BB moments and a common variance normalization; they are not simple averages of standardized BB coefficients.
- ν changes tail and boundary behavior without changing the mean-variance form.

3. Regression Models: General Structure

Consider a sample of size n:

$$\{(Y_i, m_i, \mathbf{x}_i, \mathbf{z}_i, \mathbf{u}_i) : i = 1, \ldots, n\}, \qquad Y_i \in \{0, 1, \ldots, m_i\}.$$

$$g_\mu(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad g_\delta(\delta_i) = \mathbf{z}_i^\top \boldsymbol{\alpha}, \quad g_\nu(\nu_i) = \mathbf{u}_i^\top \boldsymbol{\gamma}.$$

- $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\boldsymbol{\gamma}$ are regression coefficients.
- $\mathbf{x}_i$, $\mathbf{z}_i$, and $\mathbf{u}_i$ are covariate vectors.
- These covariate vectors may be the same, partially overlapping, or different.
- For parameters in $(0, 1)$, logit links are natural.

3.1 Binomial, BB, and B2B Regression

$$\text{Binomial:} \quad Y_i \sim \text{Binomial}(m_i, \mu_i).$$

$$\text{BB:} \quad Y_i \sim \text{BetaBinomial}(m_i, \mu_i, \delta_i).$$

$$\text{B2B:} \quad Y_i \sim \text{B2B}(m_i, \mu_i, \delta_i, \nu_i).$$

Regression moves from mean-only to distributional modeling.

3.2 Regression Log-Likelihoods

For $\mu_i = g_\mu^{-1}(x_i^\top \beta)$, the binomial log-likelihood is

$$\ell_{Bin}(\beta) = \sum_{i=1}^{n} \left[\log \binom{m_i}{y_i} + y_i \log \mu_i + (m_i - y_i) \log(1 - \mu_i) \right].$$

For BB, with

$$\alpha_i = \mu_i(1 - \delta_i)/\delta_i, \qquad \beta_i^* = (1 - \mu_i)(1 - \delta_i)/\delta_i,$$

$$\ell_{BB}(\beta, \alpha) = \sum_{i=1}^{n} \left[\log \binom{m_i}{y_i} + \log B(y_i + \alpha_i, m_i - y_i + \beta_i^*) - \log B(\alpha_i, \beta_i^*) \right].$$

For B2B,

$$\ell_{B2B}(\beta, \alpha, \gamma) = \sum_{i=1}^{n} \log \{ \text{b2b}(y_i \mid m_i, \mu_i, \delta_i, \nu_i) \},$$

$$\text{b2b}(y_i \mid m_i, \mu_i, \delta_i, \nu_i) = \int_0^1 \text{bb}(y_i \mid m_i, \mu_i, w) \, \text{be}(w \mid \delta_i, \nu_i) \, dw.$$

3.3 Model Comparison Criteria

For a fitted model with log-likelihood ℓ, p effective parameters, and sample size n:

$$\text{Deviance} = -2\ell, \qquad \text{AIC} = -2\ell + 2p,$$

$$\text{BIC} = -2\ell + p\log(n).$$

Smaller AIC and BIC indicate better penalized fit. Deviance measures fit without a complexity penalty.

Normalized randomized quantile residuals should be approximately standard normal under an adequate model (Dunn and Smyth 1996).

4. Practical Estimation: GAMLSS and Reproducibility

The notebook uses GAMLSS (Rigby and Stasinopoulos 2005; Stasinopoulos and Rigby 2007):

- `BI` for Binomial and `BB` for Beta-Binomial from `gamlss.dist` (Stasinopoulos and Rigby 2023),
- `B2B` from `RobustDiscrete` for Beta-2-Binomial (Bayes et al. 2026).

In `BB`, $\delta = \sigma/(1+\sigma)$; in `B2B`, $\sigma = \delta$.

Data and numerical differentiation use `gamlss.data` (Stasinopoulos and Rigby 2025) and `numDeriv` (Gilbert and Varadhan 2019).

Run `BoundedCountRegression.Rmd` to reproduce the simulations and figures.

The repository application is shown as code and reported results; the manuscript does not evaluate that fit.

4.1 Example: Simulated Binomial Data

Simulation settings (seed 2026):

$$n = 400, \quad m = 20, \quad x_i \sim U(-1.5, 1.5), \qquad \text{logit}(\mu_i) = -0.4 + 1.1 x_i.$$

$$Y_i \sim \text{Binomial}(20, \mu_i).$$

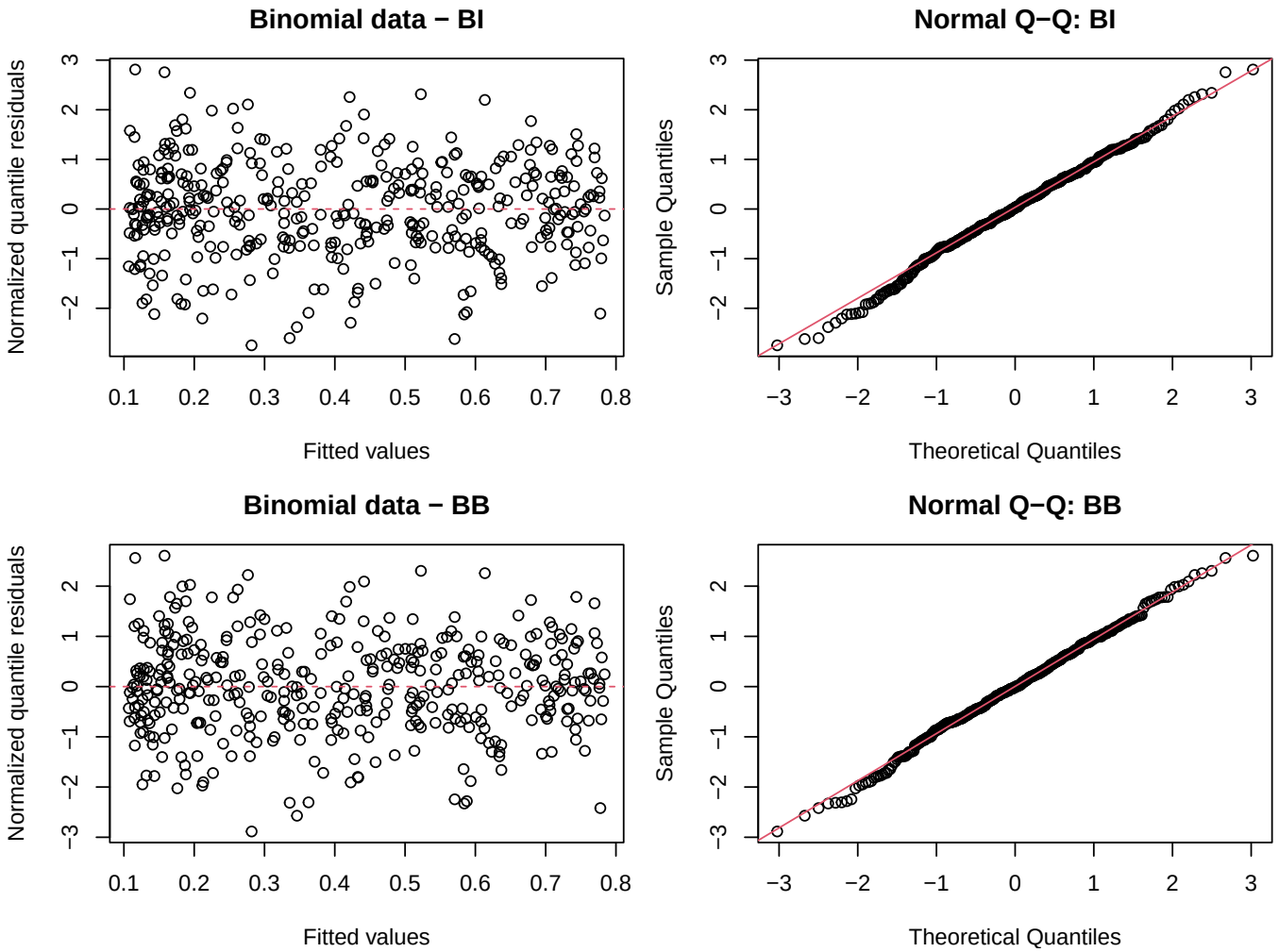
```
fit_bin_BI <- gamlss(cbind(y, failures) ~ x,
  family = BI, data = dat_bin, trace = FALSE)

fit_bin_BB <- gamlss(cbind(y, failures) ~ x,
  sigma.formula = ~ 1, family = BB,
  data = dat_bin, trace = FALSE)
```

Model	Deviance	AIC	BIC
Binomial	1623.154	1627.15	1635.14
BB	1623.155	1629.15	1641.13

The Binomial model is preferred; BB estimates essentially zero overdispersion.

4.2 Residual Diagnostics: Binomial Data



4.3 Example: Simulated Beta-Binomial Data

Simulation settings (seed 2027):

$$n = 400, \quad m = 20, \quad x_i \sim U(-1.5, 1.5), \qquad \text{logit}(\mu_i) = -0.4 + 1.1 x_i.$$

$$Y_i \sim \text{BetaBinomial}(20, \mu_i, \delta), \qquad \delta = 0.30.$$

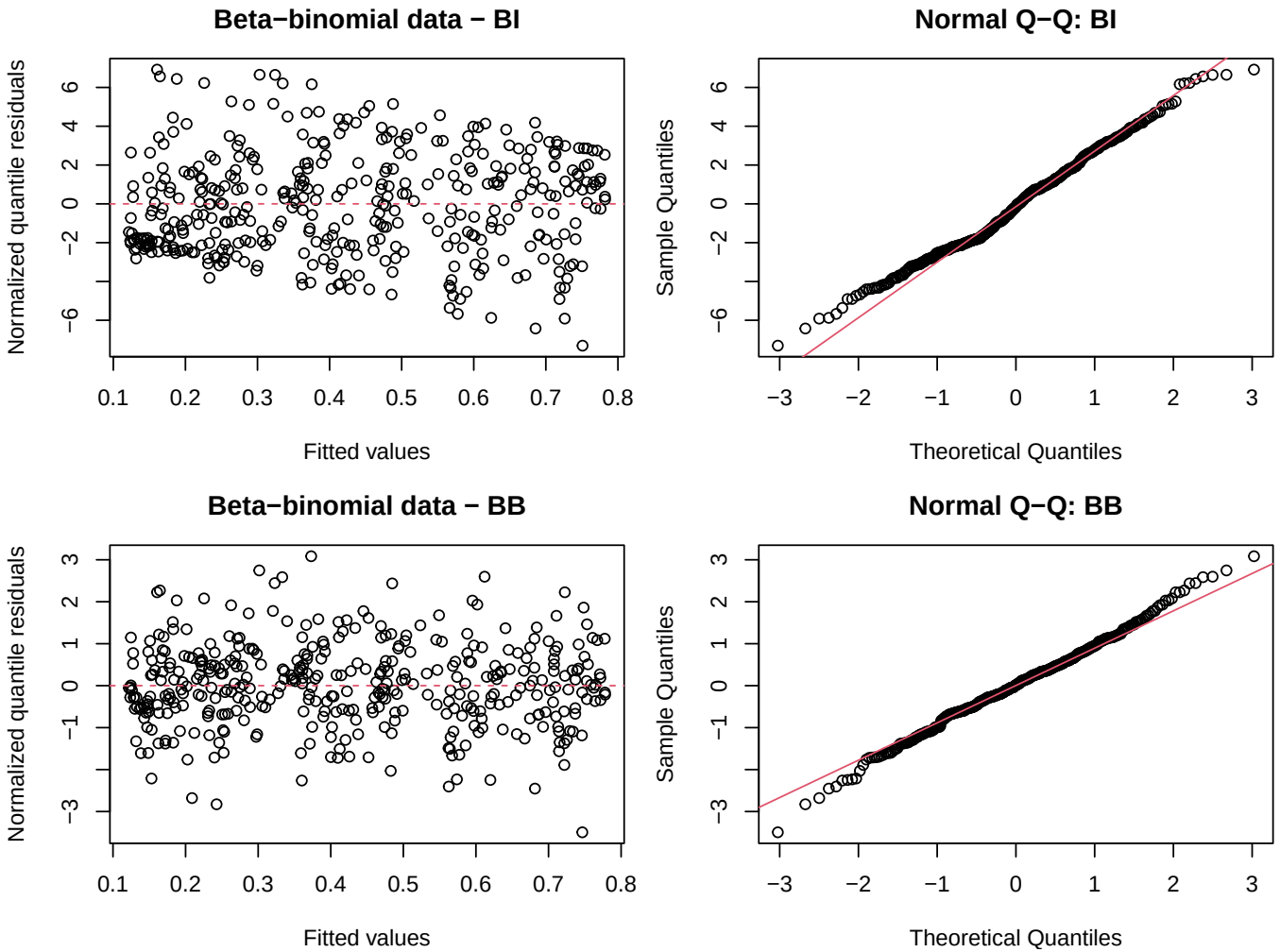
```
fit_bb_BI <- gamlss(cbind(y, failures) ~ x,
  family = BI, data = dat_bb, trace = FALSE)

fit_bb_BB <- gamlss(cbind(y, failures) ~ x,
  sigma.formula = ~ 1, family = BB,
  data = dat_bb, trace = FALSE)
```

Model	Deviance	AIC	BIC
Binomial	4013.314	4017.31	4025.30
BB	2185.892	2191.89	2203.87

The BB model is strongly preferred; $\widehat{\delta} \approx 0.318$.

4.4 Residual Diagnostics: BB Data



4.5 Example: Simulated B2B Data

Simulation settings (seed 2028):

$$n = 200, \quad m = 20, \quad x_i \sim U(-1.5, 1.5), \qquad \text{logit}(\mu_i) = -0.4 + 1.1 x_i.$$

$$Y_i \mid W_i = w_i \sim \text{BetaBinomial}(20, \mu_i, w_i), \qquad W_i \sim \text{Beta}(\delta, \nu),$$

$$\delta = 0.30, \qquad \nu = 0.60.$$

```
control_b2b <- gamlss.control(n.cyc = 100, trace = FALSE)

fit_b2b_BI <- gamlss(cbind(y, failures) ~ x,
  family = BI, data = dat_b2b, control = control_b2b)

fit_b2b_BB <- gamlss(cbind(y, failures) ~ x,
  sigma.formula = ~ 1, family = BB,
  data = dat_b2b, control = control_b2b)

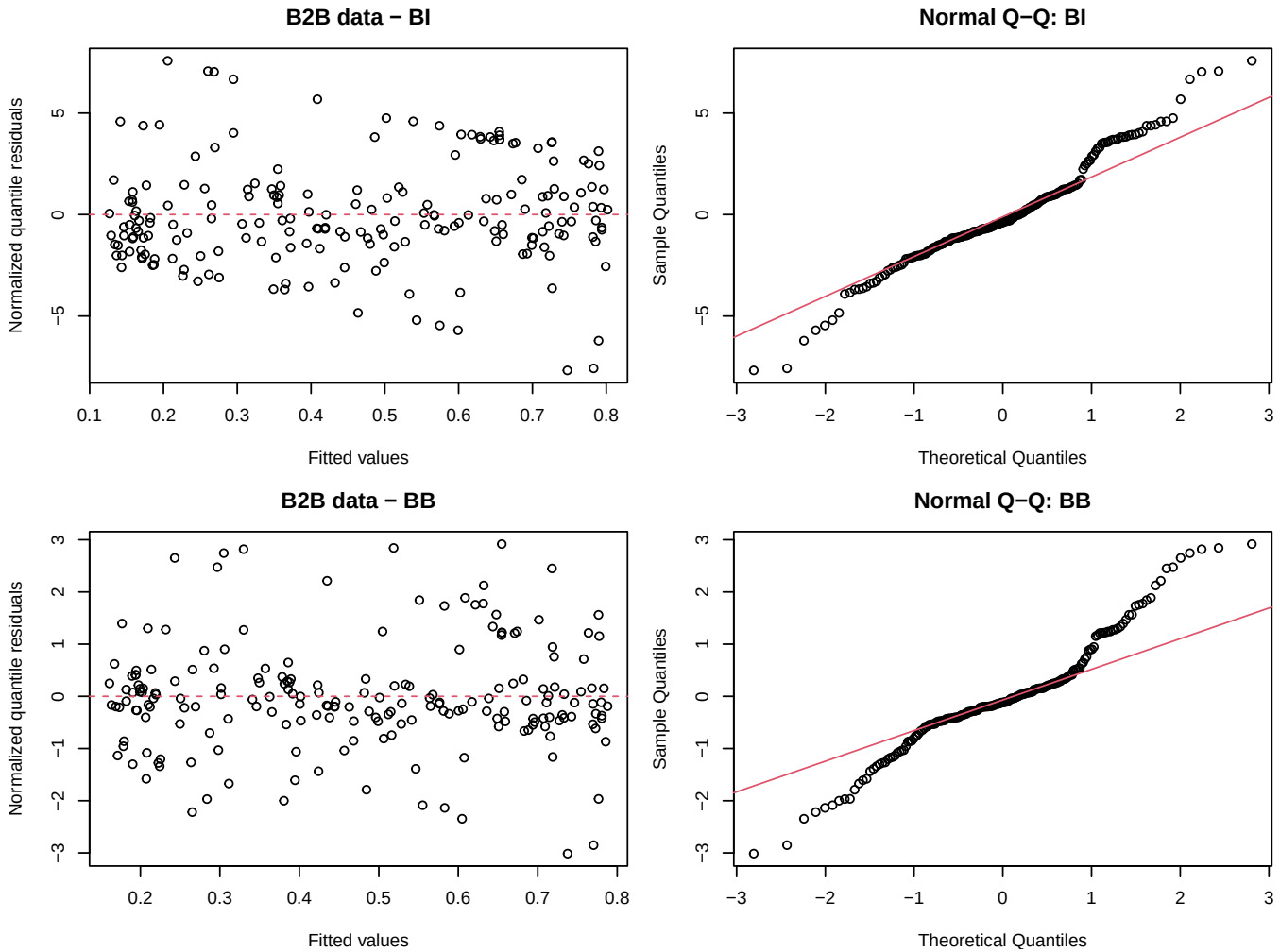
fit_b2b_B2B <- gamlss(cbind(y, failures) ~ x,
  sigma.formula = ~ 1, nu.formula = ~ 1, family = B2B,
  data = dat_b2b, mu.start = fit_b2b_BB$mu.fv,
  sigma.start = fit_b2b_BB$sigma.fv / (1 + fit_b2b_BB$sigma.fv),
  nu.start = rep(0.5, n), control = control_b2b)
```

4.6 Model Comparison: Simulated B2B Data

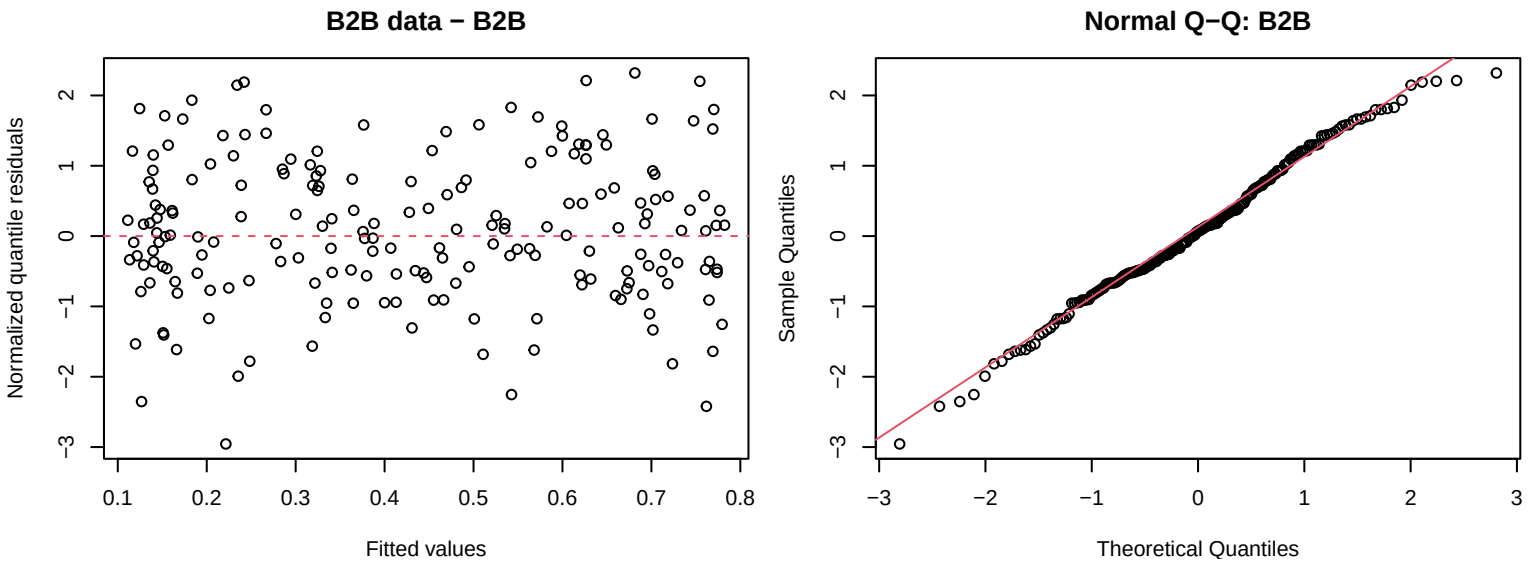
Model	Deviance	AIC	BIC
Binomial	1897.793	1901.79	1908.39
BB	1108.997	1115.00	1124.89
B2B	1013.697	1021.70	1034.89

The B2B model is preferred by all criteria.

4.7 Residual Diagnostics: BI and BB Fits



4.8 Residual Diagnostics: B2B Fit



4.9 Residual Analysis Results

Model	Mean	SD	Min	Median	Max
Binomial	-0.020	2.551	-7.677	-0.380	7.577
BB	-0.019	1.021	-3.014	-0.119	2.918
B2B	0.092	1.015	-2.956	0.069	2.319

- The Binomial residual SD is much larger than one, indicating severe unmodeled extra-binomial variation.
- The BB fit reduces the residual spread substantially.
- The B2B fit keeps the residual SD close to one and reduces the largest positive residual.
- Together with AIC and BIC, the residual analysis supports selecting the B2B model.

4.10 Regression Coefficients: Simulated B2B Data

Model	Parameter	Term	Estimate	SE	p-value
Binomial	μ	Intercept	-0.274	0.036	$<$0.001
Binomial	μ	x	1.115	0.043	$<$0.001
BB	μ	Intercept	-0.177	0.088	0.045
BB	μ	x	0.993	0.106	$<$0.001
B2B	μ	Intercept	-0.408	0.057	$<$0.001
B2B	μ	x	1.125	0.069	$<$0.001

The B2B fit recovers the generating intercept -0.4 well; its slope estimate is close to 1.1.

4.11 Dispersion and Shape: Simulated B2B Data

Model	Link scale	Term	Estimate	SE	p-value
BB	$\log \sigma$	Intercept	-0.755	0.126	$<$0.001
B2B	logit δ	Intercept	-0.971	0.148	$<$0.001
B2B	logit ν	Intercept	0.717	0.297	0.017

For BB, $\delta = \sigma/(1+\sigma)$. B2B uses logit links for δ and ν. On the B2B response scale:

$$\widehat{\delta} \approx 0.275, \qquad \widehat{\nu} \approx 0.672.$$

5. Repository Example: Beta-2-Binomial Model

Hospital-stay application (Section 5.3 and Appendix C of Bayes et al. (2024)), using `aep` from `gamlss.data` (Stasinopoulos and Rigby 2025). Code and reported results: (Bayes et al. 2026).

```
M2 <- gamlss(y ~ ward + loglos + year,
  sigma.formula = ~ year + ward,
  family = BB, data = aep)

M5 <- gamlss(y ~ ward + loglos + year,
  sigma.formula = ~ year + ward, nu.formula = ~ year,
  family = B2B, data = aep, mu.start = M2$mu.fv,
  sigma.start = M2$sigma.fv / (1 + M2$sigma.fv))
```

Reported B2B fit: deviance 4449.109, AIC 4471.109, and SBC 4528.661. This M5 B2B fit improves on the BB model M2. In the paper's wider comparison, AIC favors M5, while BIC favors M4 (constant B2B shape).

6. Exercises: Hospital Data and Software Comparison

1. Fit Binomial, BB, and B2B models to `aep` (Bayes et al. 2024; Stasinopoulos and Rigby 2025). Compare AIC, BIC, residuals, and convergence; interpret the mean, dispersion, and shape submodels.
2. Simulate BB data and compare `gamlss`/`gamlss.dist`, `aod`, and `VGAM` (Stasinopoulos and Rigby 2007; Stasinopoulos and Rigby 2023; Lesnoff and Lancelot 2012; Yee 2010, 2015). Put parameter estimates on a common scale.

6. Exercises: Additional Applications

3. Use `aod::orob1` or `aod::orob2` to compare Binomial, BB, and B2B models (Lesnoff and Lancelot 2012). Assess fitted means, fit criteria, and randomized quantile residuals.
4. Use `VGAM::sdac` to compare the same model families with common mean covariates (Yee 2010, 2015). Explain differences in dispersion parameterization.
5. Implement a Bayesian approach for the models studied here.

7. Conclusions

- Binomial regression is the essential baseline.
- BB regression is preferred when extra-binomial variation is present.
- B2B regression is useful when overdispersion is accompanied by tail or boundary behavior.
- Covariates may be included in μ, δ, and ν, but identification and stability must be checked.
- Model choice should combine AIC, BIC, normalized quantile residuals, and substantive interpretation.
- `BoundedCountRegression.Rmd` reproduces the derivations, simulations, model outputs, and figures.

8. References I

Bayes, C. L., J. L. Bazán, and L. Valdivieso. 2024. "A Robust Regression Model for Bounded Count Health Data." *Statistical Methods in Medical Research* 33 (8): 1392–411. https://doi.org/10.1177/09622802241259178.

Bayes, Cristian L., Jorge L. Bazán, and Luis Valdivieso. 2026. *RobustDiscrete Installer*. GitHub repository. https://github.com/cbayesr/RobustDiscrete_installer.

Dunn, Peter K., and Gordon K. Smyth. 1996. "Randomized Quantile Residuals." *Journal of Computational and Graphical Statistics* 5 (3): 236–44. https://doi.org/10.1080/10618600.1996.10474708.

Gilbert, Paul, and Ravi Varadhan. 2019. *numDeriv: Accurate Numerical Derivatives*. https://doi.org/10.32614/CRAN.package.numDeriv.

Johnson, N. L., A. W. Kemp, and S. Kotz. 2005. *Univariate Discrete Distributions*. 3rd ed. Wiley.

8. References II

Lesnoff, Matthieu, and Renaud Lancelot. 2012. *Aod: Analysis of Overdispersed Data*. https://doi.org/10.32614/CRAN.package.aod.

Rigby, R. A., and D. M. Stasinopoulos. 2005. "Generalized Additive Models for Location, Scale and Shape." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 54 (3): 507–54. https://doi.org/10.1111/j.1467-9876.2005.00510.x.

Stasinopoulos, D. M., and R. A. Rigby. 2007. "Generalized Additive Models for Location Scale and Shape (GAMLSS) in R." *Journal of Statistical Software* 23 (7): 1–46. https://doi.org/10.18637/jss.v023.i07.

Stasinopoulos, Mikis, and Robert Rigby. 2023. *Gamlss.dist: Distributions for Generalized Additive Models for Location Scale and Shape*. https://doi.org/10.32614/CRAN.package.gamlss.dist.

Stasinopoulos, Mikis, and Robert Rigby. 2025. *Gamlss.data: Data for Generalized Additive Models for Location Scale and Shape*. https://doi.org/10.32614/CRAN.package.gamlss.data.

8. References III

Yee, Thomas W. 2010. "The VGAM Package for Categorical Data Analysis." *Journal of Statistical Software* 32 (10): 1–34. https://doi.org/10.18637/jss.v032.i10.

Yee, Thomas W. 2015. *Vector Generalized Linear and Additive Models: With an Implementation in R*. Springer. https://doi.org/10.1007/978-1-4939-2818-7.